Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/29/2004
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	HZE Radiation: Modulation Of Genetic Effects By RNA Interference Of NHEJ		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/01/2003	**End Date:**	05/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: end date changed to 5/31/2004 because new grant with same title awarded from solicitation 03-OBPR-07 in 2004 (jvp 1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bedford, Joel (Colorado State University) Liber, Howard (Colorado State University)		
Grant/Contract No.:	NAG9-1516		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	For extended missions in space, astronauts will be exposed to a radiation environment unlike any on the earth's surface. Galactic cosmic rays include a particularly worrisome component of energetic iron nuclei which are high atomic number, high energy (HZE) charged particles with linear energy transfer (LET) similar to alpha particles but with very different track structures. While the relative biological effectiveness (RBE) of a particular radiation roughly depends on the LET, the very different track structures for these two radiations may well modulate this RBE even if they have identical average LETs. Further, this "radiation quality" for the exposures actually relevant for astronauts will be modified by interactions with materials in the spacecraft, including their own bodies. Basic to virtually all relevant biological effects of ionizing radiation is the underlying damage produced in DNA, and the subsequent cellular processing of such damage. The damage can be qualitatively different for different kinds of radiations, and the genetics of the biological systems exposed can greatly affect damage processing and ultimate outcome—the biological effect of concern. The accurate repair of DNA double-strand breaks (DSB) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosome rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. It is important, therefore, to understand cellular responses to DNA damage and to elucidate the mechanisms and phenotypic consequences of DSB misrepair. Our general aim in this proposal is to examine connections between cellular radiosensitivity phenotypes relevant for carcinogenesis, namely chromosome aberration formation, mutagenesis and telomere fusions, and deficiencies in various components of the non-homologous end joining system, using RNA interference to systematically knockdown gene products known to be involved, either singly or in combination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In 2002, we proposed to utilize the relatively new technique of RNA interference (RNAi) to deplete levels of various components of the non-homologous end joining (NHEJ) double-strand break (DSB) repair system, and to study the effects on HZE radiation-induced chromosomal rearrangements and gene mutation. We were awarded one-year of support. The suggestion from reviewers and NASA program officers was that before the project could be considered for further funding, it would be necessary to demonstrate that the proposed studies involving siRNA knockdown of various proteins would be feasible. During this year (June 1, 2003 - May 31, 2004) we have collected considerable additional data supporting our ability to conduct theses studies. The accurate repair of DNA double-strand breaks (DSBs) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosomal rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. We have focused on the consequent phenotypic effects of faulty repair by examining connections between cellular radiosensitivity phenotypes relevant for carcinogenesis after exposure to IR, and deficiencies in various components of the non-homologous end joining (NHEJ) system. We produced deficiencies of individual components of the DNA-dependent protein kinase (DNA-PK) holoenzyme (Ku86 and the catalytic subunit, DNA-PKcs), both singly and in combination, using RNA interference (RNAi) in human lymphoblastoid cell lines. Exposure of cells exhibiting reduced protein expression to either gamma-rays or 1GeV/nu Fe particles demonstrated differential effects on telomere dysfunction and mutation frequency, as well as differential effects between radiation qualities. Our initial results are reported in the article "Suppression of DNA-PK by RNAi has different quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma-rays or HZE particles" by Qinming Zhang, Eli S. Williams, Kristin F. Askin, Yuanlin Peng, Joel S. Bedford, Howard L. Liber and Susan M. Bailey, currently in press (2005) in Radiation Research (NASA supplement). With continued support, our experiments will lead to valuable information regarding CPR question 10.12, “What are the effects of inter-individual diversity?” We hypothesize that NHEJ repair components are major genetic determinants controlling individual susceptibility to radiation-induced cancer, as well as heritable effects of exposure. Therefore, naturally occurring deficiencies of these proteins within the population will put these individuals at increased risk. We also anticipate addressing CPR question 10.07, “What are the cancer risks in humans from spaceflight?” We anticipate that our characterization of telomere dysfunction, chromosome aberration formation and gene mutation will serve as surrogate endpoints for predicting cancer risks.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Zhang Q, Williams ES, Askin KF, Peng Y, Bedford JS, Liber HL, Bailey SM. "Suppression of DNA-PK by RNAi has different quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma rays or HZE particles." Radiation Research, in press, May 2004. , May-2004